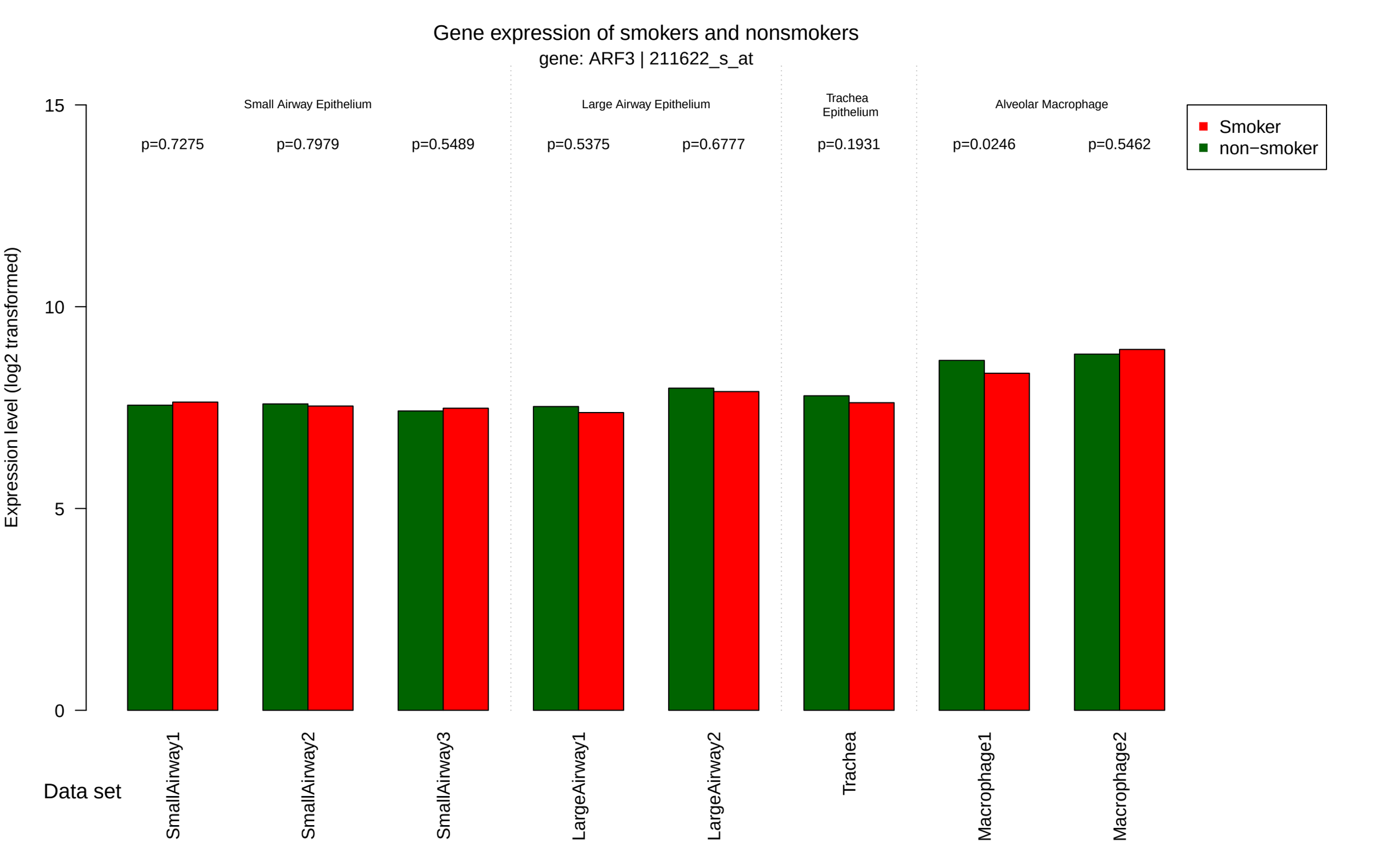
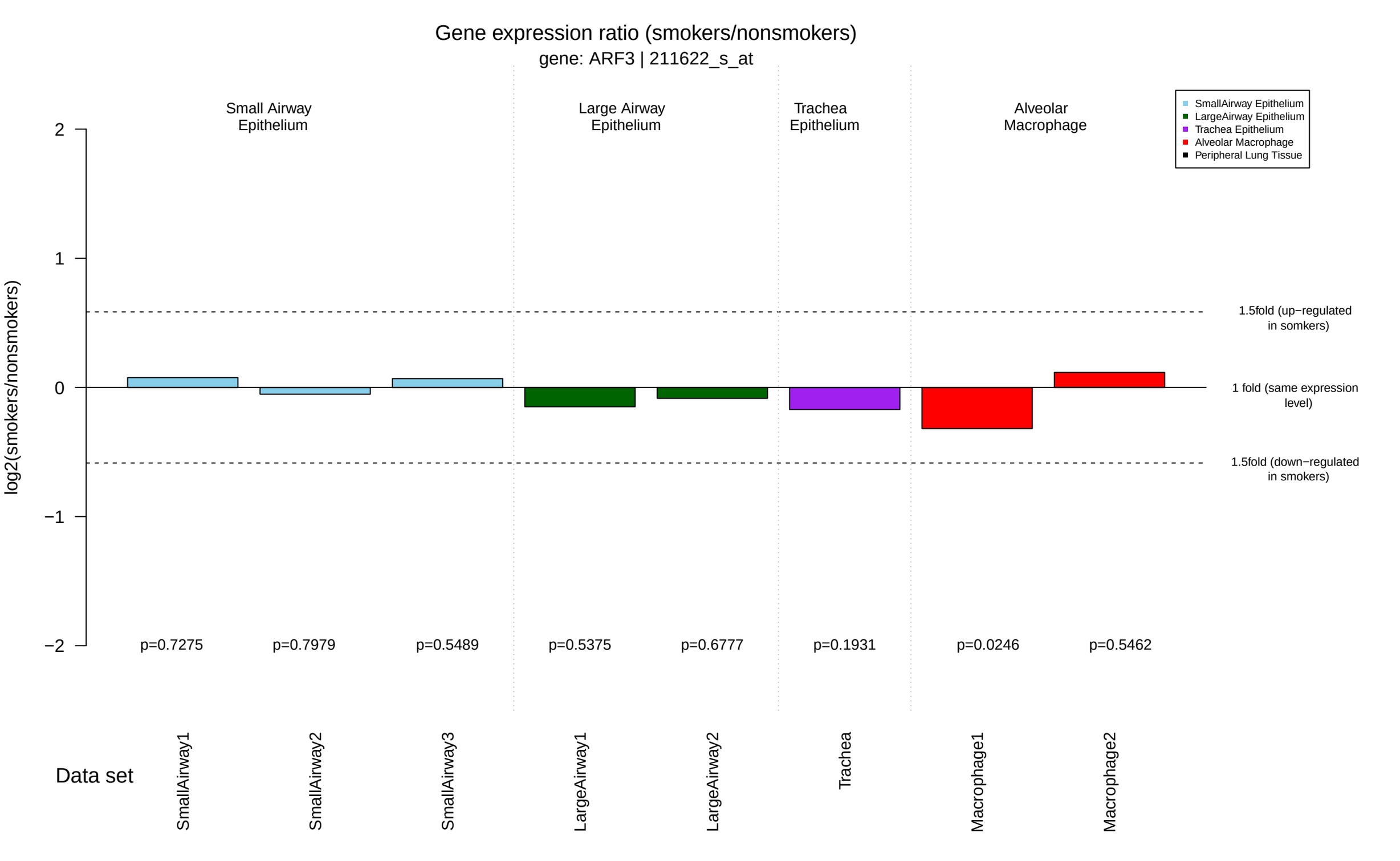
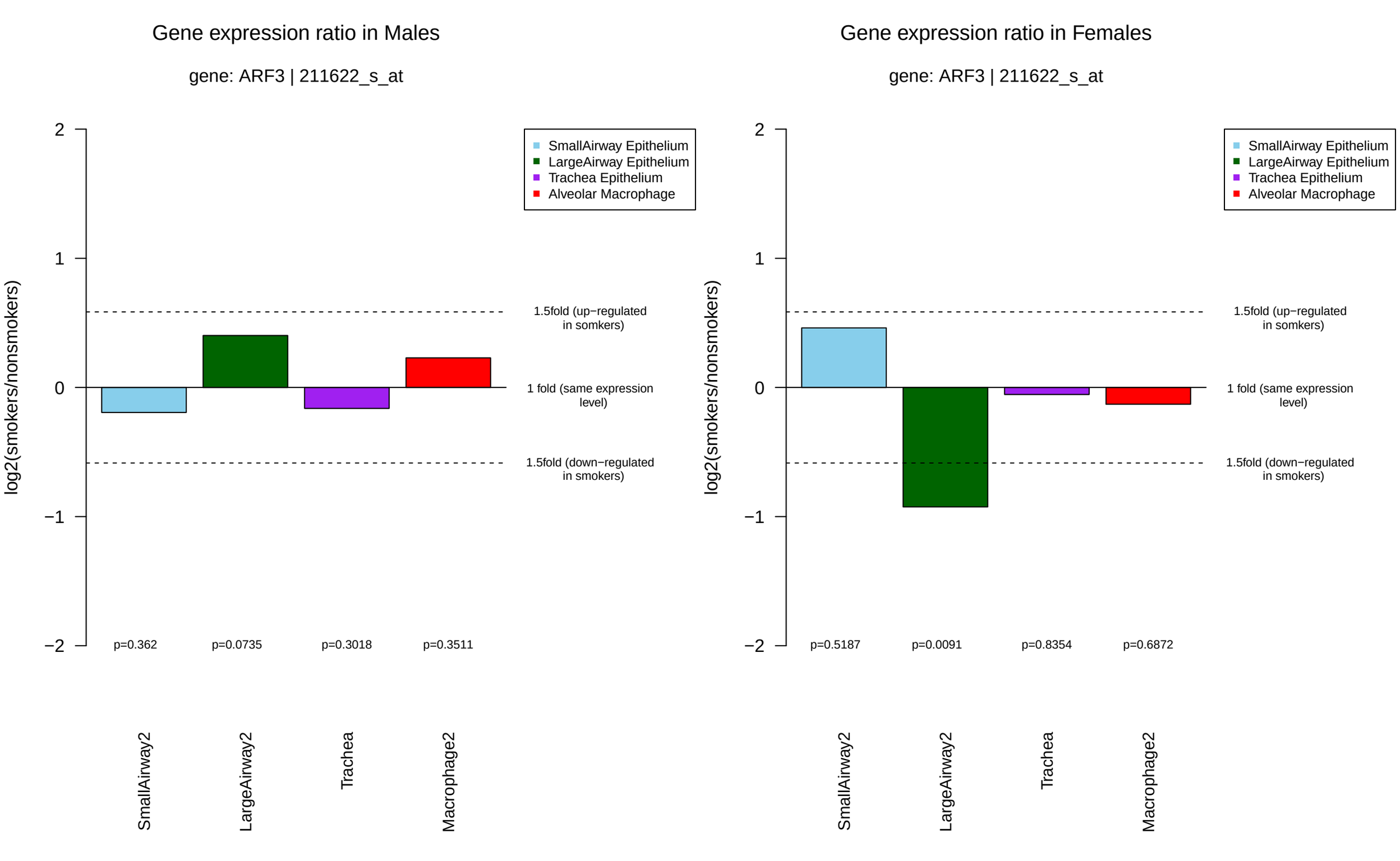
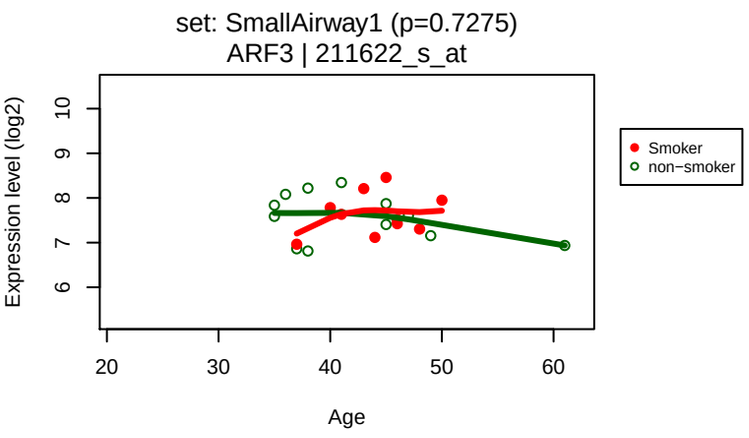
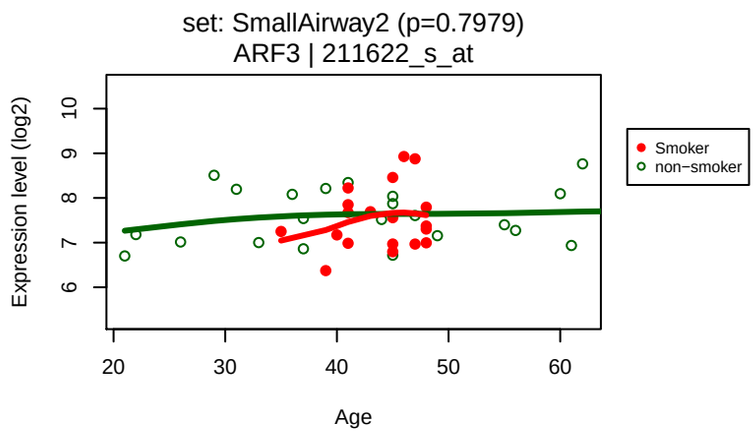
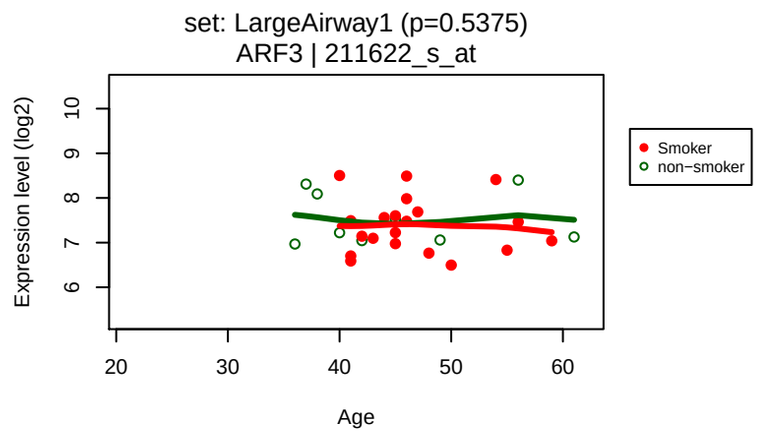

Gene expression of smokers and nonsmokers
gene: ARF3 | 211622_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7275
p=0.7979
p=0.5489
p=0.5375
p=0.6777
p=0.1931
p=0.0246
p=0.5462
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

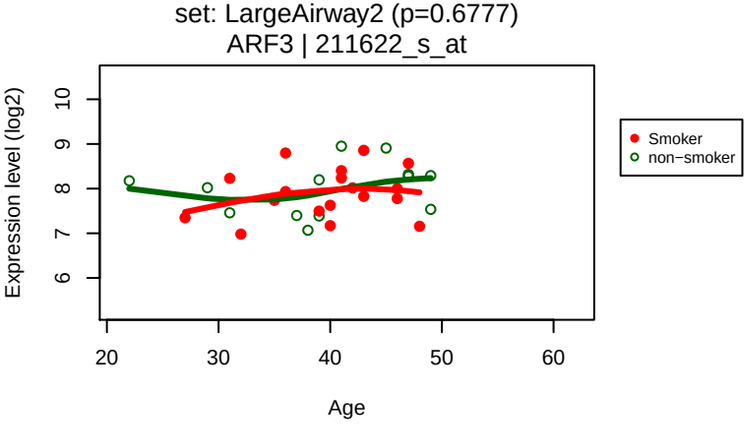

Gene expression ratio (smokers/nonsmokers)
gene: ARF3 | 211622_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7275
p=0.7979
p=0.5489
p=0.5375
p=0.6777
p=0.1931
p=0.0246
p=0.5462
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

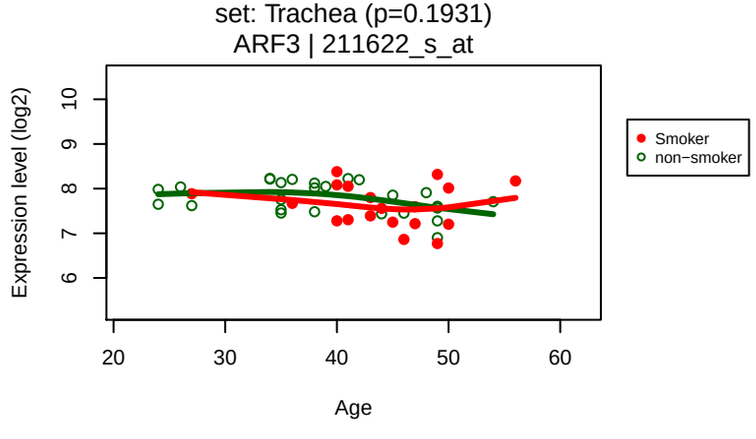

Gene expression ratio in Males
gene: ARF3 | 211622_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.362
p=0.0735
p=0.3018
p=0.3511
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARF3 | 211622_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5187
p=0.0091
p=0.8354
p=0.6872
SmallAirway2
LargeAirway2
Trachea
Macrophage2

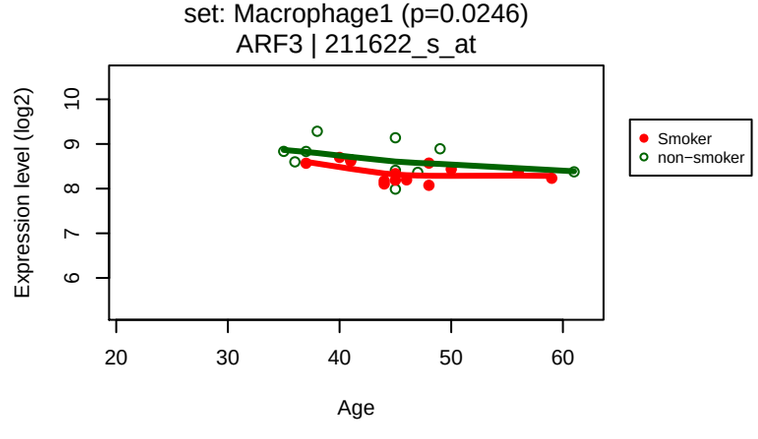

set: SmallAirway1 (p=0.7275)
ARF3 | 211622_s_at
Expression level (log2)
Age
Smoker
non−smoker

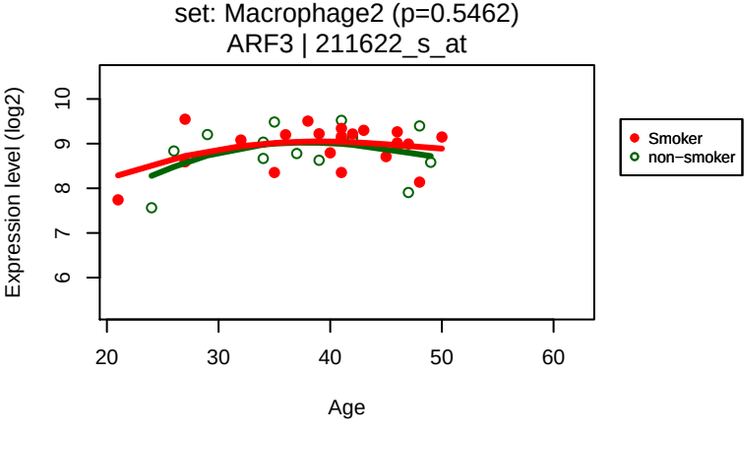

set: SmallAirway2 (p=0.7979)
ARF3 | 211622_s_at
Expression level (log2)
Age
Smoker
non−smoker

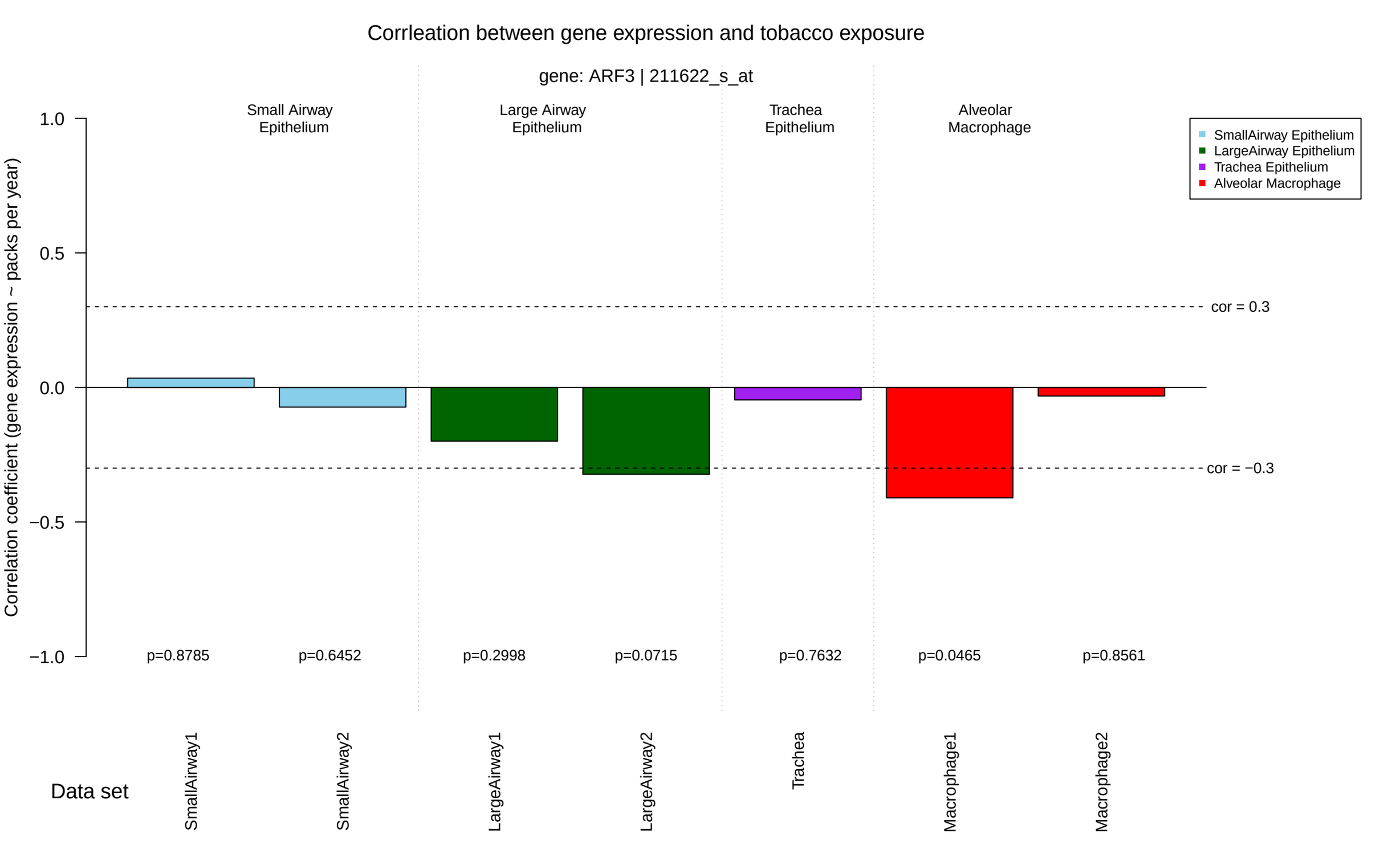

set: LargeAirway1 (p=0.5375)
ARF3 | 211622_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6777)
ARF3 | 211622_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1931)
ARF3 | 211622_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0246)
ARF3 | 211622_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5462)
ARF3 | 211622_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARF3 | 211622_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8785
p=0.6452
p=0.2998
p=0.0715
p=0.7632
p=0.0465
p=0.8561
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2